SENTIMENTAL ANALYSIS OF PRODUCT REVIEWS

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INTRODUCTION

In the era of social media dominance, sentiment analysis has emerged as a critical tool for extracting public opinions and driving market strategies. This project leverages sentiment analysis on Twitter data to analyze customer opinions about products across specific locations. By targeting localized outcomes, our approach provides actionable insights for businesses, enabling efficient allocation of marketing resources. This research demonstrates the utility of sentiment analysis for both technical and non-technical domains, highlighting its potential to shape data-driven decisions in the modern marketplace.

The exponential growth of social media platforms has transformed how people share opinions, making platforms like Twitter a goldmine for sentiment analysis. Traditional survey methods fail to capture the spontaneous and authentic feedback that social media offers. This project uses sentiment analysis techniques to analyze product reviews shared on Twitter, focusing on sentiments across specific geographic locations. By identifying these trends, businesses can refine their strategies, optimizing efforts in areas of low sentiment and sustaining popularity in high-sentiment regions.

PAPER OVERVIEW

PAPER 1 - **Localized Twitter opinion mining using sentiment analysis**

Overview

This paper explores methodologies for sentiment analysis using Twitter data, focusing on specific locations to understand product-related opinions. The study employs several Natural Language Processing (NLP) tools, including Stanford NLP for data preprocessing, VADER for sentiment scoring, and NamSor for gender classification. The primary objective is to extract marketing insights by identifying geographic and gender-specific preferences for product features.

Applications

The analysis targets national and location-specific trends in public sentiments about the iPhone 6. Positive feedback highlights the camera's performance, while negative sentiments focus on issues like the screen's durability and the infamous "bend issue." These insights provide valuable input for tailored marketing strategies and product feature enhancements.

Findings

The paper demonstrates the effectiveness of localized Twitter opinion mining using sentiment analysis techniques, offering actionable insights for geographic-specific marketing and product development.

PAPER 2- **Sentiment Analysis Using Product Review Data**

Overview

Paper investigates sentiment analysis using machine learning techniques, focusing on sentiment polarity categorization at sentence and review levels. It compares classifiers such as Naïve Bayes and Support Vector Machines (SVM).

Applications

The research examines brand perception and customer satisfaction using data from Facebook and Twitter. These insights can help improve customer engagement strategies and refine sentiment analysis models.

Findings

Random Forest performed best for sentence level categorization. SVM and naive bayes showed better result in Naive Bayes.

OUR APPROACH

For our project, we combined methodologies from two research papers to create a comprehensive analysis.

From Paper 1, we were inspired to focus on user reviews for specific products (e.g., iPhones). We used the **VADER** sentiment analysis tool to classify these reviews as **Positive or Negative.** Additionally, we visualized the sentiment distribution across different countries and product variants to gain insights into geographical and model-specific trends.

From Paper 2, we adopted user ratings for exploratory data analysis (EDA). Specifically, we leveraged the ratings column in our dataset to identify trends, such as average ratings by country and product variants. This helped us explore user preferences and uncover broader patterns. Furthermore, we implemented a **Wordcloud** to identify the most frequently used words in the reviews. Building on this, we extended the Wordcloud analysis to distinguish between positive and negative words in reviews. We then analyzed these words further to determine which product features received more positive or negative sentiment.

This comprehensive analysis enables us to provide actionable insights for the company, helping them identify areas for improvement and better align with user preferences.

METHODOLOGY

- **Data Collection**: Used iPhone_reviews dataset from Kaggle.
- **Data Preprocessing**: Remove punctuation, numbers, and tokenization of text and poorly structured sentences to ensure data reliability.
- **Sentiment Analysis**: Used VADER for sentiment scoring to classify text reviews into positive and negative
- **Localization**: Grouped reviews by geographic location and variants to identify sentiment trends regionally and also across different variants.
- **Result Interpretation:** Analyze and visualize data trends and sentiment trends arosss countries and variants. Also used word cloud analysis to understand which feature are driving more positive or negative sentiment.

TOOLS AND LIBRARIES

Pandas, Numpy, scikit-learn, Matplotlib, Seaborn, Wordcloud, NLTK, collections.counter(count word frequencies in text data), VADER (for sentiment classification)

ALGORITHMS AND TECHNIQUES

- **Classification Algorithm:** SVM, Random Forest, Logistic Regression, Naive Bayes for training model.
- **Word Frequency Analysis:** Counts occurrences of words in positive and negative reviews using Counter to identify key trends in sentiment data.
- **Word Cloud Generation:** Visualizes the frequency of top words in reviews, helping in exploratory sentiment analysis.

IMPLEMENTATION DETAILS

DATASET

For our project **"iPhone_reviews"** dataset is taken from Kaggle. The dataset has 3000+ reviews posted by users on Amazon across 7 countries for a period of 3 years (2021 to 2023). Reviews also have colors and storage information.

Column Name	Description
productAsin	Unique Product ID
Country	The country from which the review is posted
Date	Date of review
isVerified	Whether the review is verified or not
ratingScore	Ratings posted (1 to 5)
reviewTitle	Title of Review
reviewDescription	Review Text
Colors	Review posted for which color
Sizes	Review posted for which storage option

Insights

1. 7 countries (India, Japan, United States, UAE, Egypt, Canada, Mexico)
2. 9 colors (Black, Blue, Green, Midnight, Pink, Purple, Red, Starlight, Yellow)
3. 3 storage options (128GB, 256GB, 512GB)
4. The reviews are in the period from Fall 2021 to Fall 2024
5. Ratings on a scale of 1 to 5

```
  productAsin country        date  isVerified  ratingScore              reviewTitle  \
0  B09G9BL5CP   India 2024-08-11        True            4                No charger
1  B09G9BL5CP   India 2024-08-16        True            5             iPhone 13 256GB
2  B09G9BL5CP   India 2024-05-14        True            4  Flip camera option nill

                                   reviewDescription    Colors   Sizes
0  Every thing is good about iPhones, there's not...  Midnight  256 GB
1  It look so fabulous, I am android user switche...  Midnight  256 GB
2  I tried to flip camera while recording but no ...  Midnight  256 GB
```

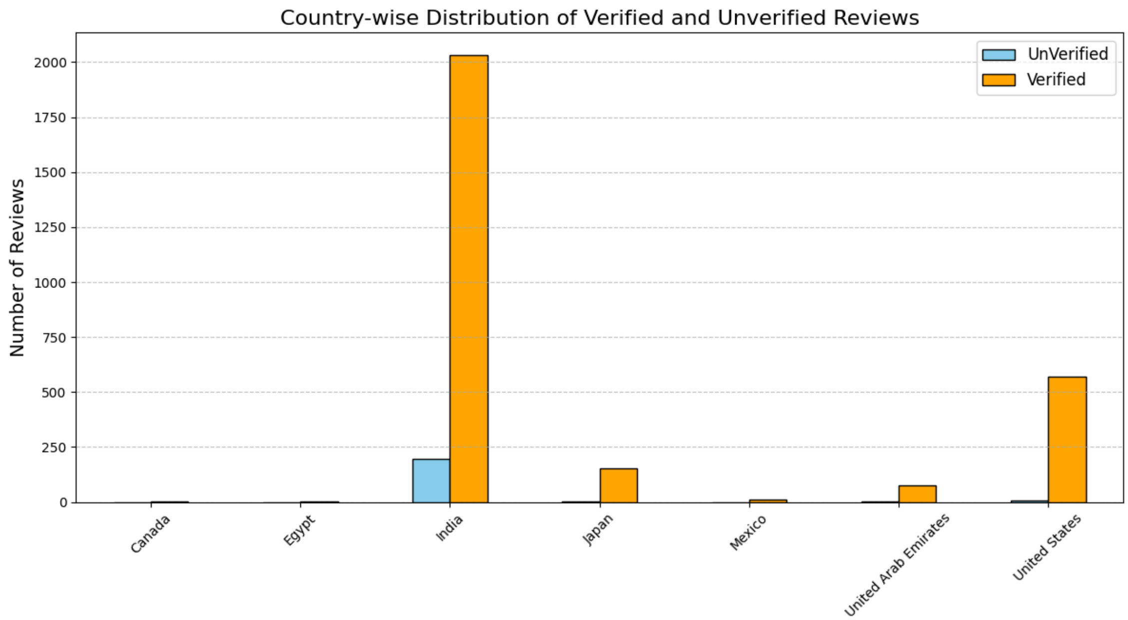
	0	1
productAsin	7	[B09G9BL5CP, B09P82T3PZ, B09G9J5JZX, B0CHX1W1X...
country	7	[India, Japan, United Arab Emirates, Egypt, Un...
date	789	[2024-08-11 00:00:00, 2024-08-16 00:00:00, 202...
isVerified	2	[True, False]
ratingScore	5	[4, 5, 3, 2, 1]
reviewTitle	2018	[No charger, iPhone 13 256GB, Flip camera opti...
reviewDescription	2296	[Every thing is good about iPhones, there's no...
Colors	9	[Midnight, Pink, Blue, Green, Starlight, Red, ...
Sizes	3	[256 GB, 128 GB, 512 GB]

DATA PREPROCESSING

Missing Values Check: We see missing values for reviewDescription column but we decided not to remove it as these rows still have reviewTitle and we will later combine these columns to generate one column

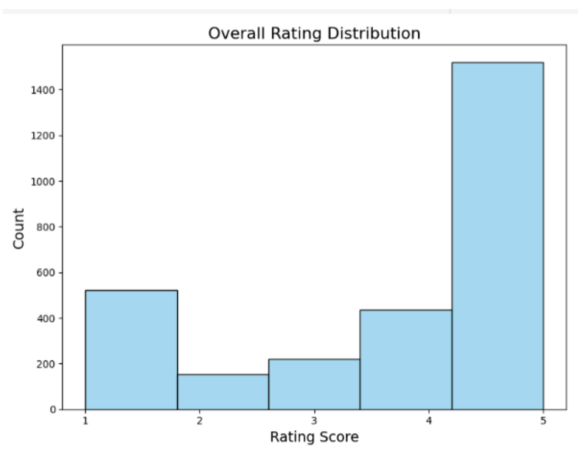
```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3062 entries, 0 to 3061
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   productAsin        3062 non-null   object
 1   country            3062 non-null   object
 2   date               3062 non-null   datetime64[ns]
 3   isVerified         3062 non-null   bool
 4   ratingScore        3062 non-null   int64
 5   reviewTitle        3062 non-null   object
 6   reviewDescription  2975 non-null   object
 7   Colors             3062 non-null   object
 8   Sizes              3062 non-null   object
dtypes: bool(1), datetime64[ns](1), int64(1), object(6)
memory usage: 194.5+ KB
```

Remove UnVerified Reviews: We noticed there are 212 unverified reviews. We decided to remove these to generate correct insights. Most of the unverified reviews were from India



DATA VISUALIZATION

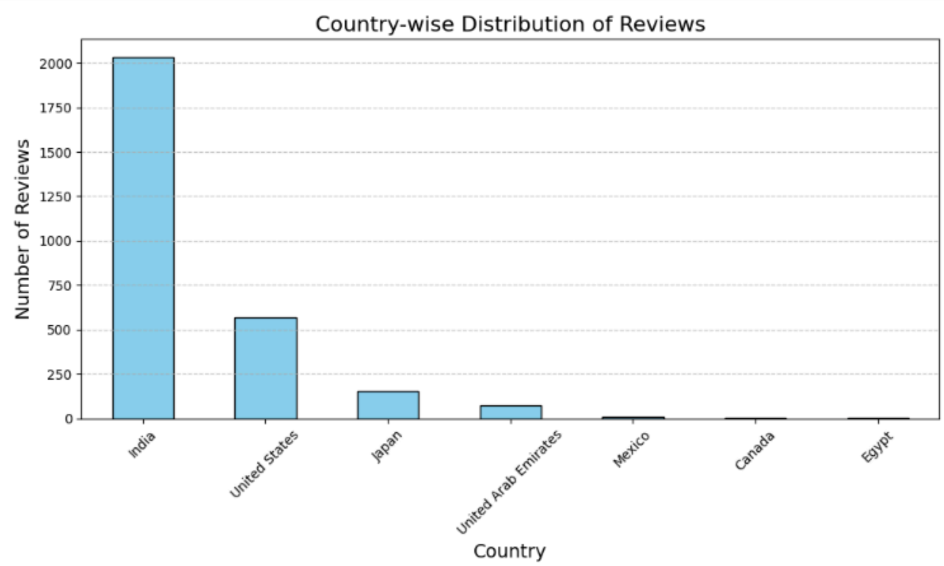
- **Overall Rating Distribution:** Ratings are **not** fairly distributed:



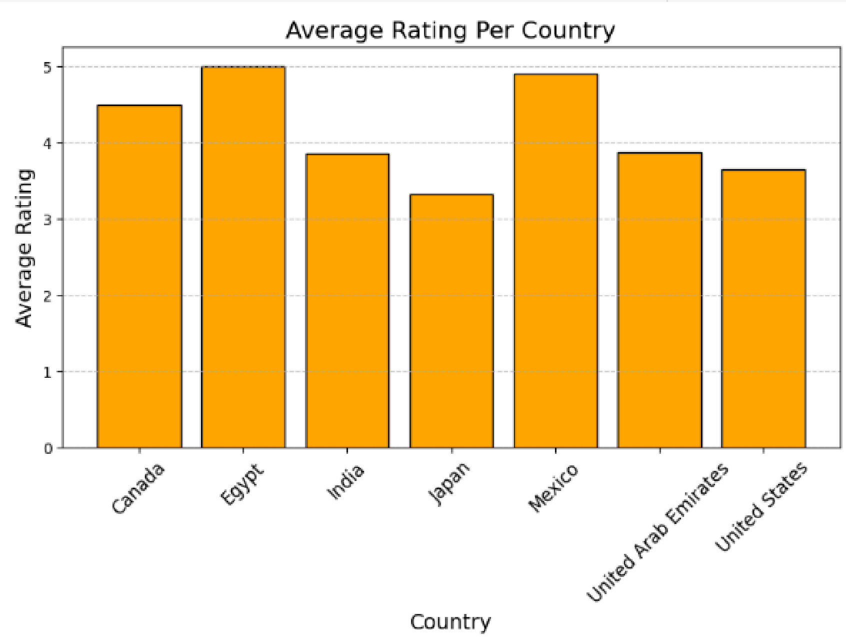
- Most ratings are 5 indicating overall user satisfaction
- Followed by 1's: This is interesting as brands like Apple getting 1's could be a matter to explore

1. **Country-wise distribution of reviews and average rating per country**

India and the US are major contributors, contributing to more than 80% of reviews. This could mean that iPhone is more popular in these countries compared to others.

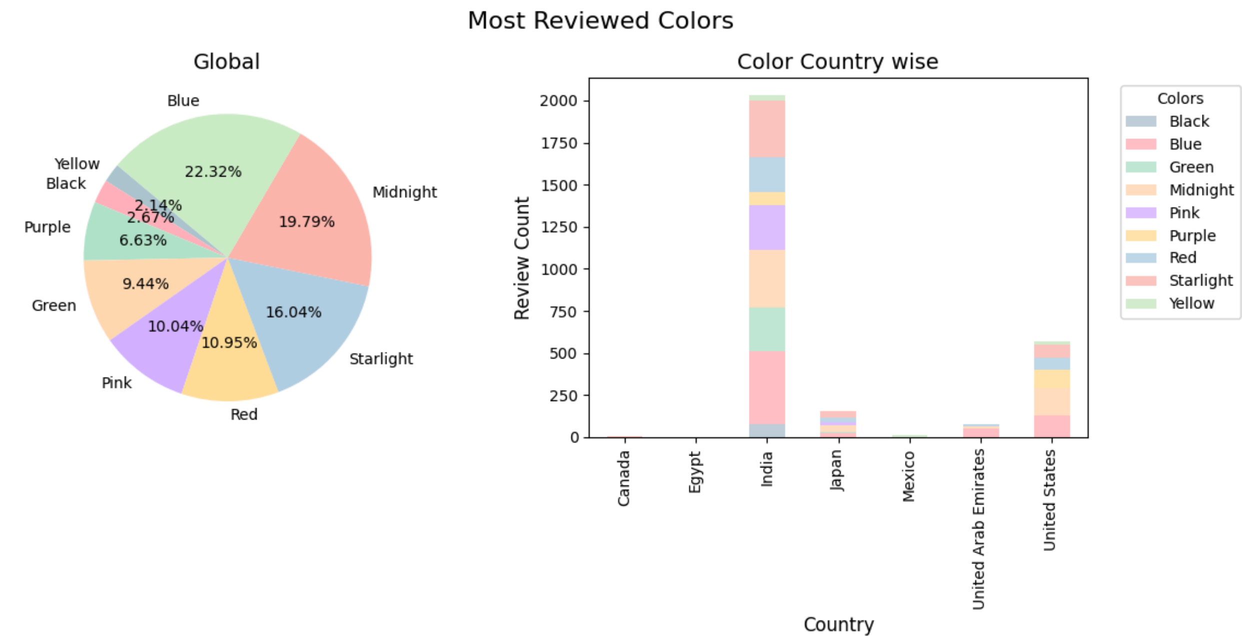


India has an overall 3.8 rating according to 2000+ reviews. United States has a 3.7 rating from 500 reviews. Globally 70% of reviews are greater than 4

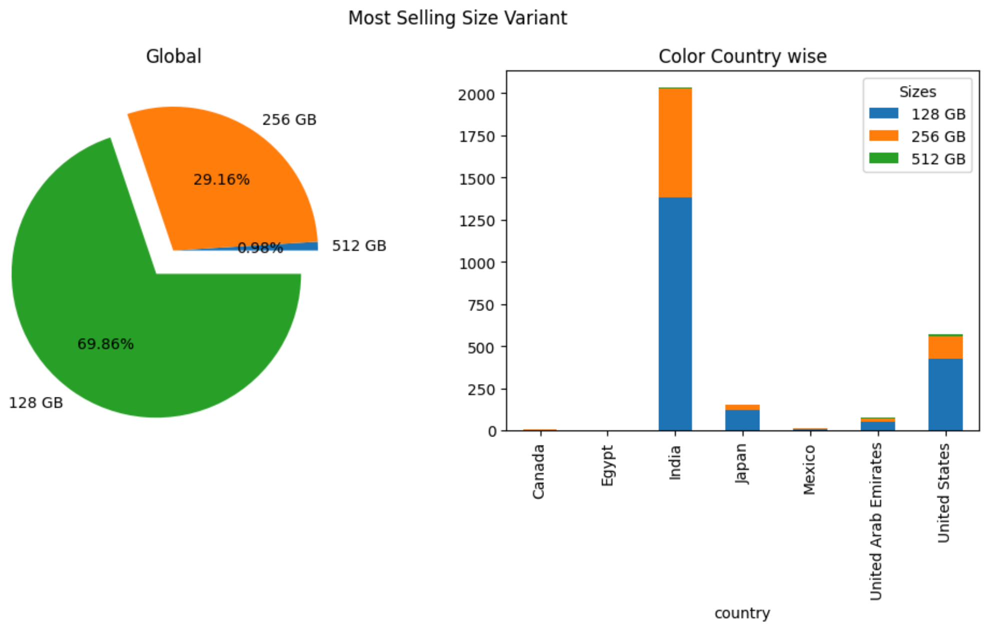


2. **Most Reviewed Colors / Sizes / Variants**

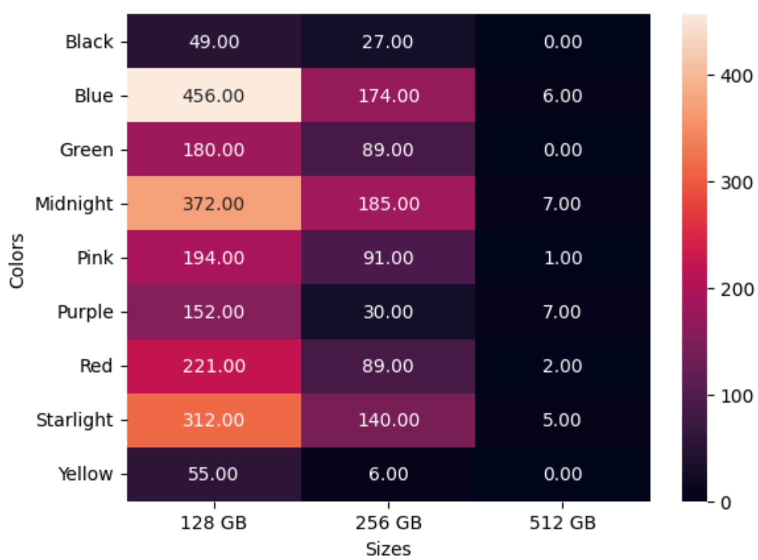
Colors: Blue (22%) and Midnight(19%) are the most reviewed colors, followed by starlight (16%). India is the leading contributor for almost all colors except Purple which has more reviews than the USA. This could indicate that the purple color is more favored in the US.



Variants: 128 GB storage has 70% of reviews and it's commonly preferred above other sizes in all countries. So it is most sold and popular choice among users. 512 GB is the least favored.



Variants: 128GB variant is the most popular choice (max number of reviews) with: 'midnight', 'blue', and 'starlight' colors.



SENTIMENTAL ANALYSIS

We combined the 'reviewTitle' and 'reviewDescription' columns into a single column 'reviewText' and dropped these two columns

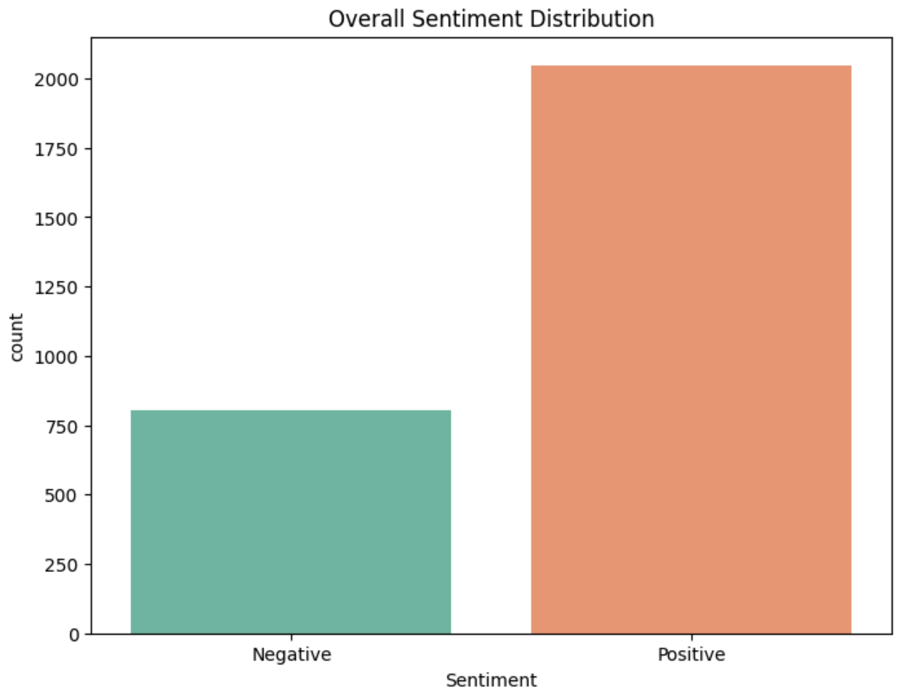
	productAsin	country	date	isVerified	ratingScore	Colors	Sizes	Variant	reviewText
0	B09G9BL5CP	India	2024-08-11	True	4	Midnight	256 GB	Midnight - 256 GB	No charger Every thing is good about iPhones, ...
1	B09G9BL5CP	India	2024-08-16	True	5	Midnight	256 GB	Midnight - 256 GB	iPhone 13 256GB It look so fabulous, I am andr...
2	B09G9BL5CP	India	2024-05-14	True	4	Midnight	256 GB	Midnight - 256 GB	Flip camera option nill I tried to flip camera...
3	B09G9BL5CP	India	2024-06-24	True	5	Midnight	256 GB	Midnight - 256 GB	Product 100% genuine
4	B09G9BL5CP	India	2024-05-18	True	5	Midnight	256 GB	Midnight - 256 GB	Good product Happy to get the iPhone 13 in Ama...

Assigned Sentiment Scores: **VADER** was used to calculate the sentiment score for each review text and, then we created a new Column '**Sentiment**' in our data frame and assigned the value as Negative or Positive based on the scores. Review Text with a score greater than 0 was classified as positive, else negative. Here's a code snippet

```
#import and initialise vader
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer
analyzer = SentimentIntensityAnalyzer()

#Create a function to assign sentiment labels
def get_sentiment(review):
    score = analyzer.polarity_scores(str(review))['compound']
    return 'Positive' if score > 0 else 'Negative'

#Applying the function to the reviewDescription column
iphone_reviews['Sentiment'] = iphone_reviews['reviewText'].fillna('').apply(get_sentiment)
```

	productAsin	country	date	isVerified	ratingScore	Colors	Sizes	Variant	reviewText	Sentiment
0	B09G9BL5CP	India	2024-08-11	True	4	Midnight	256 GB	Midnight - 256 GB	No charger Every thing is good about iPhones, ...	Negative
1	B09G9BL5CP	India	2024-08-16	True	5	Midnight	256 GB	Midnight - 256 GB	iPhone 13 256GB It look so fabulous, I am andr...	Positive
2	B09G9BL5CP	India	2024-05-14	True	4	Midnight	256 GB	Midnight - 256 GB	Flip camera option nill I tried to flip camera...	Negative
3	B09G9BL5CP	India	2024-06-24	True	5	Midnight	256 GB	Midnight - 256 GB	Product 100% genuine	Negative
4	B09G9BL5CP	India	2024-05-18	True	5	Midnight	256 GB	Midnight - 256 GB	Good product Happy to get the iPhone 13 in Ama...	Positive

71% of reviews were found positive, which means users are mostly satisfied with the product. We will explore later to analyze which factors can be contributors of negative reviews that the company can look into to increase sales and user satisfaction.



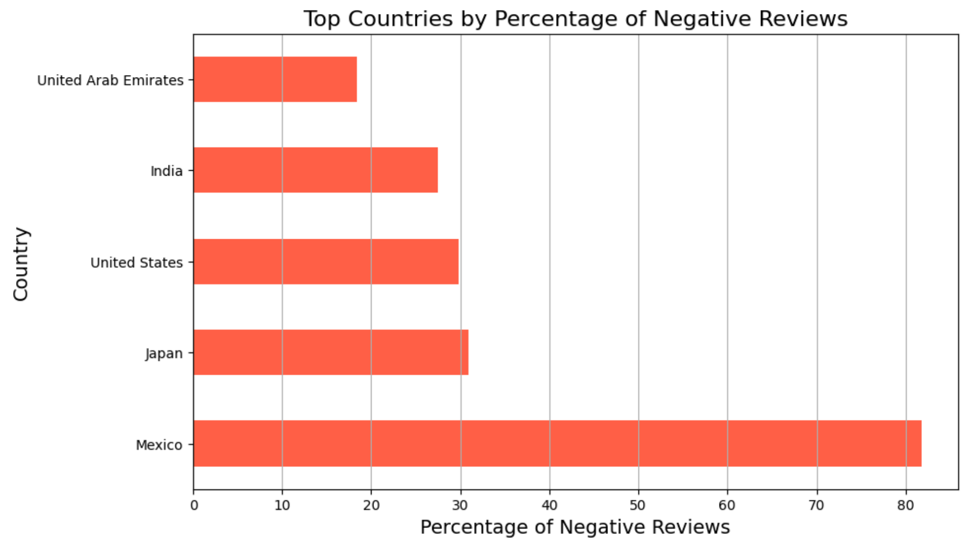
INTERPRETATION

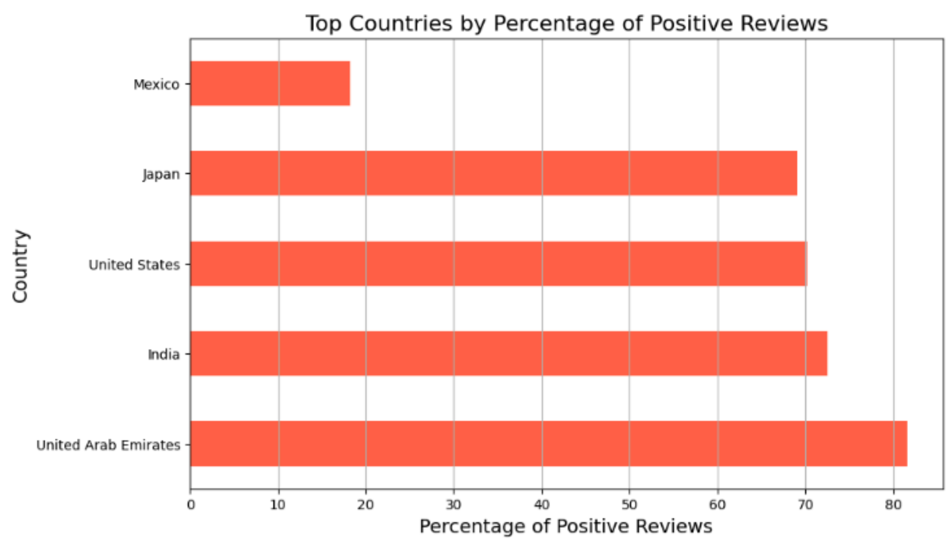
1. **Which country had the most negative and positive reviews?**

Mexico has the highest percentage of **negative reviews** (>80%) while the UAE (80%) has the highest percentage of positive reviews.

Although India was leading in the total number of reviews, the **UAE** has more percentage of **positive** reviews compared to India

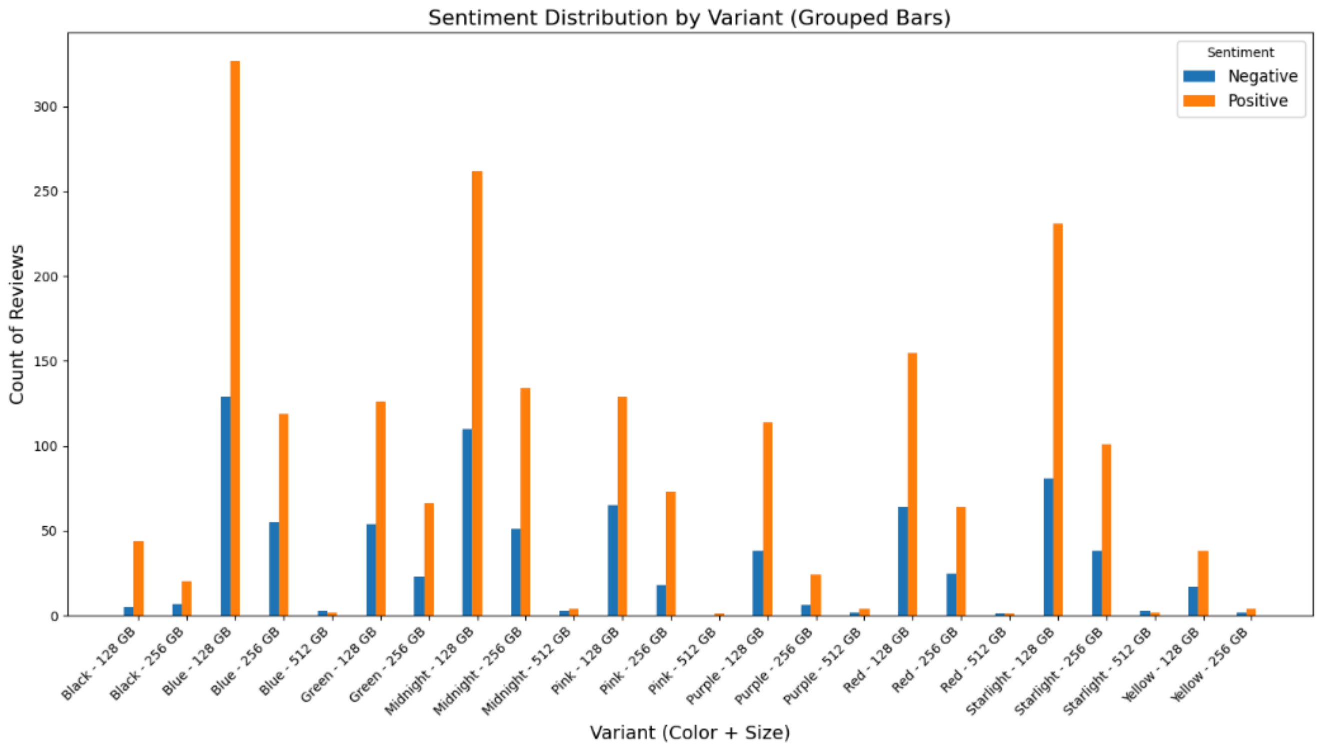
*Reviews from Egypt and Canada were dropped plotting this graph), as they were very less and we focused on countries that are major contributors.





2. Sentiment Distribution By Variant

Across most variants, positive reviews greatly outnumber negative reviews, suggesting overall customer satisfaction. Few variants have more noticeable counts of negative reviews, such as Blue - 256 GB/ Green -128GB, and Yellow -128GB. The company could investigate specific complaints further to address the issues.



TEXT PROCESSING USING NLP

Here ‘reviewText’ column was cleaned using the NLTK library to make it fit for wordCloud analysis and model training.

Steps:

1. Converted Text to Lowercase
2. Removed Punctuations
3. Removed Numbers
4. Tokenized Text
5. Remove Stopwords (eg : and, etc, or , any)

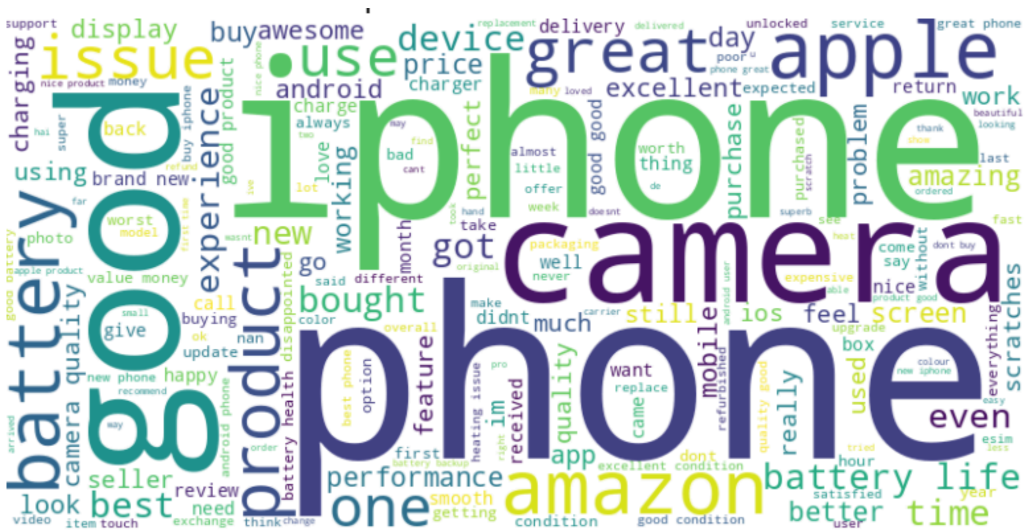
reviewText was converted to lowercase, punctuation, and numbers removed. The text was then tokenized, and after that, stopwords were removed, and the text was joined back to form a full review.

VISUALIZING SENTIMENTS THROUGH WORDCLOUD

A **word cloud** is a visual representation of the most frequently occurring words in a text dataset. The size of each word corresponds to its frequency, making it easier to identify prominent themes or patterns in the text.

For this analysis, I used word cloud visualizations to explore the most common words in text reviews reviews.The steps included:

1. **Generating an Overall Word Cloud**: All reviews were combined into a single text, and a word cloud was generated to visualize the most frequently occurring words across all reviews.

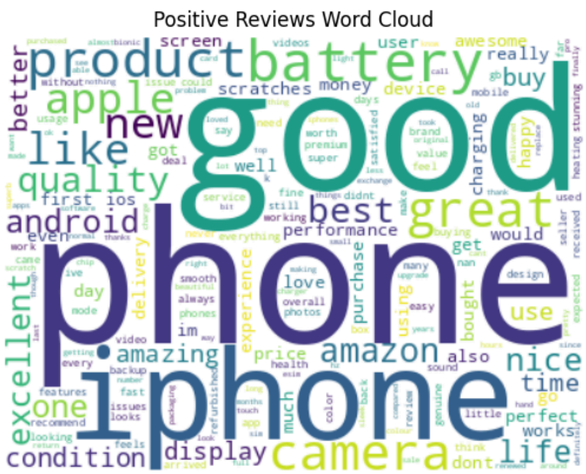


Findings: Camera, Battery, Charging, Price, Performance, Display, Working as some of words we can see. We will figure out more to see which feature have positive and negative sentiment.

2. **Positive and Negative Sentiment Word Clouds**: Reviews were segregated based on sentiment (positive and negative). Using word frequency counts, separate word clouds were created for each sentiment category, highlighting distinct trends in positive and negative feedback.

Insights:

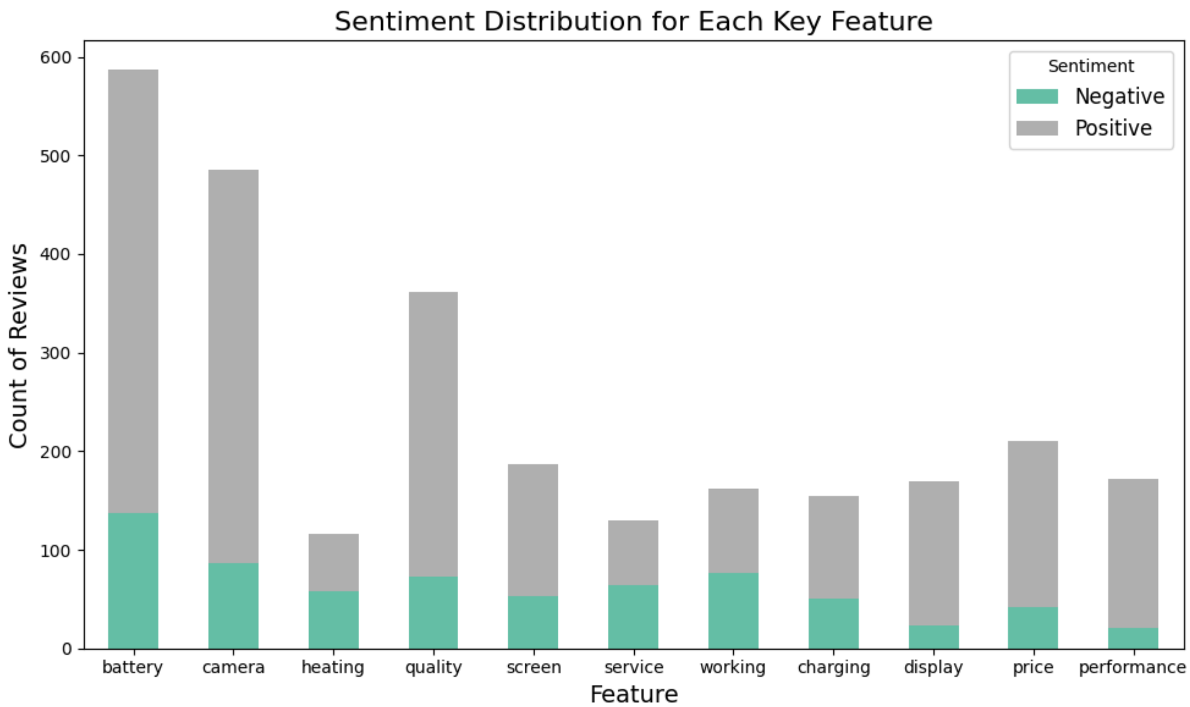
1. Reviews on **battery / Camera / Quality** are more **positive** than negative
2. We can see some **negative reviews** on related to **working**
3. There are **negative reviews** on **screen, heating and service**



We thought to analyse further and the count of most appearing words/features in overall/positive/negative cloud to understand which feature is driving more positive or negative sentiment.

We grouped together below features and plotted count of positive and negative sentiment for each of them

```
features = ['battery', 'camera', 'heating', 'quality','screen', 'service',
'working','charging','display','price','performance']
```

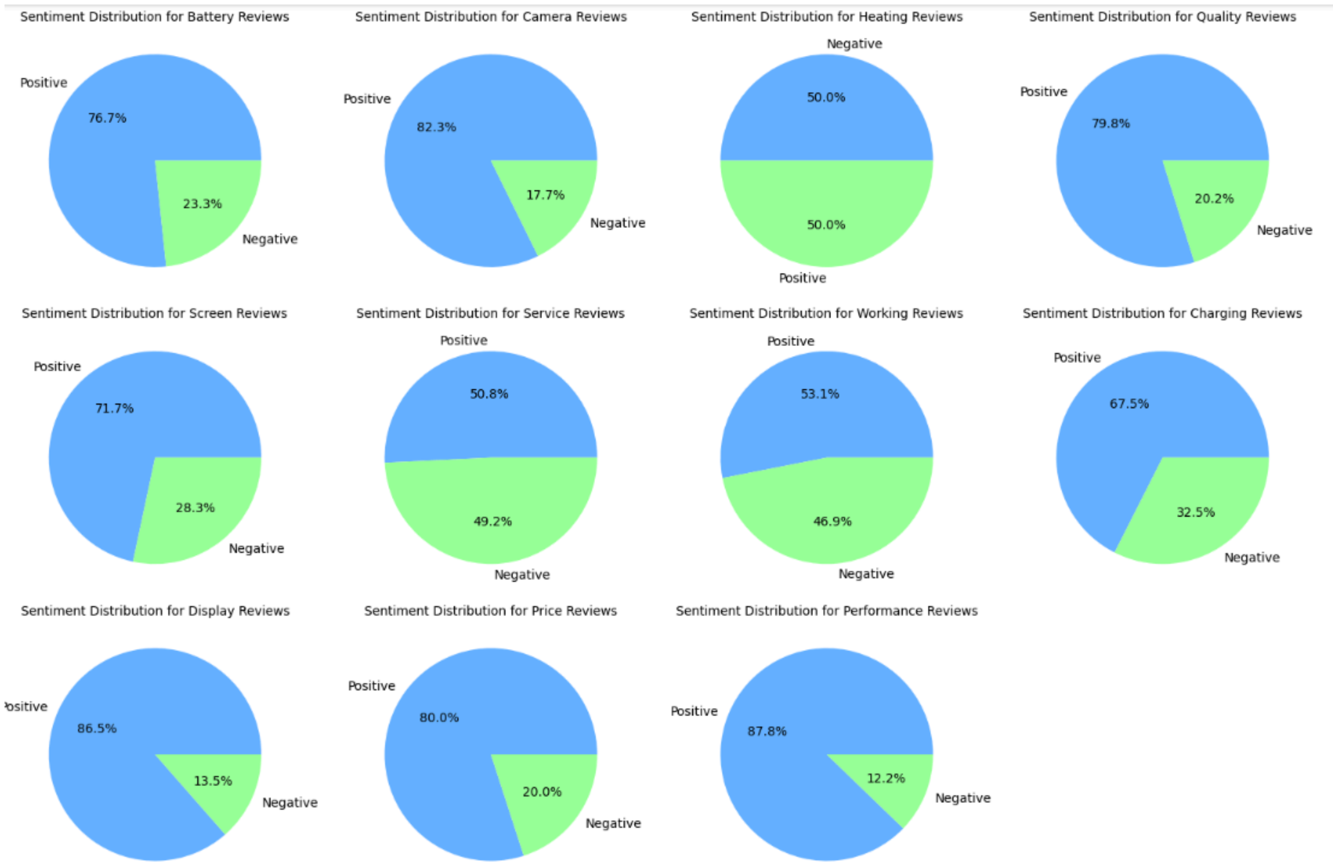


Findings: Most of features have more sentiment on positive side indicating overall customer satisfaction whereas **heating/working/charging/service** have **mixed reviews which** indicate that that there is a problem related to these taht need to be addresses.

Lets plot a pie chart to see percentage of positive and negative sentiment of these features to have more understanding:

Findings:

- **Highest percentage of positive reviews: Camera (82%)** has highest percentage followed by performance (87%) , display(86%) , quality(79%) and battery(76%)
- **Highest percentage of negative reviews: Heating(50%)** has highest value followed by Working(46%) and Service (49%)



MODEL TRAINING

We tried classification different algorithm to train : Logistic Regression, Random Forest, Naive Bayes, SVM.

SVM generated the **highest accuracy (88%)** of all of them.

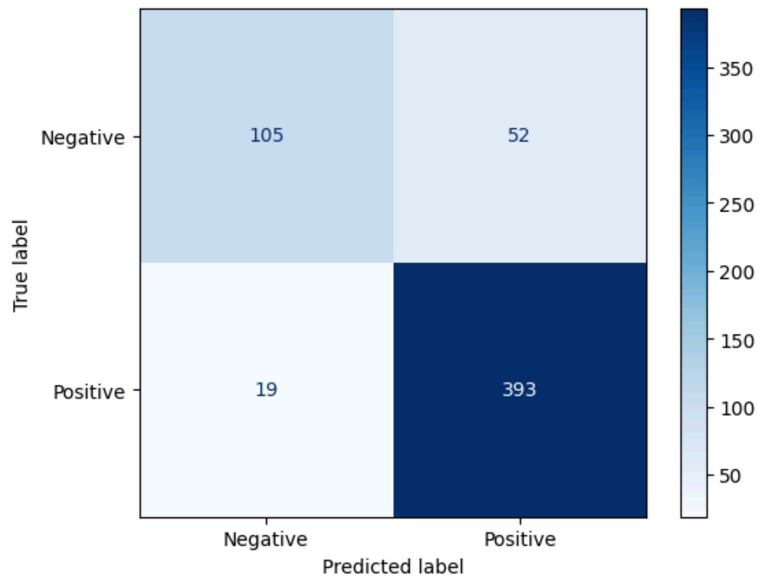
Input feature is ‘cleaned_review’ text and output is ‘SentimentEncoded’ column (label encoder was used to label positive sentiment as 1 and negative as 0)

```
#Split Data
X = iphone_reviews['cleaned_review']
y = iphone_reviews['SentimentEncoded']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

VALIDATION

Accuracy and Confusion Matrix

Model was able to correctly predict 393 positive sentiment and 105 negative sentiment. 19 positive were incorrectly identified as negative (but its a decent count). 52 negative sentiment were classified as positive.



Accuracy Score

```
Accuracy: 0.875219683655536
Classification Report:
              precision    recall  f1-score   support

    Negative       0.85      0.67      0.75       157
    Positive       0.88      0.95      0.92       412

    accuracy                           0.88       569
   macro avg       0.86      0.81      0.83       569
weighted avg       0.87      0.88      0.87       569
```

CONCLUSION

This research successfully demonstrates the potential of sentiment analysis on product reviews data as a superior alternative to traditional survey-based opinion mining. By analyzing location-specific, demographic-based, and feature-specific sentiment trends, the project provides practical insights that businesses can leverage to streamline their marketing strategies. The results highlight the versatility and relevance of this approach across various industries.

For future work, efforts will focus on improving sentiment prediction accuracy by leveraging advanced language models such as BERT. Additionally, expanding the scope of sentiment analysis to include diverse data sources, such as reviews and discussions from other social media platforms, will enhance the robustness and applicability of the findings.

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